

Methods		CIFAR100-B0									CIFAR100-B50									ImageNet100-B0		
		5 steps			10 steps			20 steps			2 steps			5 steps			10 steps			10 steps		
		2000	1000	0	2000	1000	0	2000	1000	0	2000	1000	0	2000	1000	0	2000	1000	0	2000	1000	0
DER (?)	Avg	76.80	73.67	39.93	75.36	71.10	26.53	74.09	70.34	16.96	74.61	72.50	43.89	73.21	70.17	23.31	72.81	69.67	13.11	93.23	78.06	29.15
	Last	68.32	62.22	17.31	65.22	56.77	8.88	62.48	53.32	4.80	68.84	64.70	21.45	73.21	59.99	8.85	65.45	56.98	4.80	87.52	53.96	9.96
TCIL	Avg	77.72	75.64	63.92	77.30	74.13	56.57	75.11	71.85	46.74	76.42	75.17	65.78	74.88	74.34	63.18	73.72	71.91	47.52	94.17	93.66	84.05
	Last	69.58	67.00	53.08	66.41	62.87	39.91	63.54	59.07	27.11	71.91	70.38	54.85	74.88	67.90	45.39	66.36	64.79	255.4	88.84	86.04	76.82

Table 4: Accuracy comparison with different memory budget (number of exemplars). We report the top-1 accuracy on CIFAR100 and the top-5 accuracy on ImageNet100.

DEA	Div Loss	CR	MLKD	FFM	Avg
					61.84
✓					69.45
✓	✓				70.59
✓	✓	✓			73.43
✓	✓	✓	✓		75.80
✓	✓	✓	✓	✓	77.30

Table 5: Ablations on the components of TCIL

Methods	Error	Incremental tasks									
		1	2	3	4	5	6	7	8	9	10
baseline	WTC	75	133	169	164	191	169	226	182	207	254
	ONC	0	162	450	750	856	1034	1597	1834	2053	2943
	ITC	0	0	67	237	524	794	824	1200	1481	1538
CR	WTC	75	132	173	169	199	194	232	228	248	271
	ONC	0	170	390	583	541	515	739	743	647	832
	ITC	0	0	75	291	670	1023	1249	1733	2254	2718
TCIL	WTC	75	120	157	177	212	211	223	221	238	248
	ONC	0	141	284	382	309	345	538	476	453	709
	ITC	0	0	92	309	630	879	1156	1619	2046	2391

Table 6: Error statistics on CIFAR100-B0. ITC, ONC, WTC denote inter-task confusion, old-new confusion, within-task confusion, respectively.

tion of the total number of errors, less than 10% at the last step. This result indicates that ITC and ONC are the main cause of catastrophic forgetting, which is different from the case of fixed network structure that often suffers from weight drift (??). Our exploration basically verifies that to address the catastrophic forgetting in DEAs, more attention needs to be paid to the task confusion, especially ITC and ONC.

We set up a controlled trial with DER to illustrate TCIL is much less dependent on rehearsal. It is clearly shown in Tab. 4 that as the memory size decreases, the gap between TCIL and DER in accuracy becomes larger no matter the dataset, evaluation protocols and task splits. The result strongly demonstrates that TCIL proposed in this paper can alleviate the problem that DER relies much on the rehearsal mechanism under the structure of dynamic extension.

Conclusion

We have inspected class incremental learning from the task confusion angle, where ITC and ONC have been pointed out as two major causes of catastrophic forgetting in DEAs. As a consequence, TCIL is presented. It develops a multi-level knowledge distillation to promote knowledge propagation and utilization, while attention mechanism and classifier re-scoring strategy are also taken into account. The ex-

periments conducted on CIFAR100 and ImageNet100 basically verify our proposal. TCIL and TCIL-lite consistently report state-of-the-art accuracy. They are also more robust with the accumulation of incremental steps and less sensitive to the availability of rehearsal memory. The error statistics imply that task confusion is largely reduced by TCIL especially ONC errors. The strength of error reduction, though, is still limited by the fact that the knowledge in different sub-networks is still not being enough shared and fused. Thus, future work includes the exploration of more effective knowledge sharing and utilization protocols. Meanwhile, we are also interested in extending TCIL to other research scenarios, e.g., the label sets are not exclusive.

Acknowledgments

This project was supported by National Key R&D Program of China (No. 2021ZD0112804) and in part by the National Natural Science Foundation of China (No. 62172103, 62102092)

Amet nisi soluta, quos sit recusandae culpa corporis molestias accusantium quis a ullam aut, perferendis pariatur laboriosam facilis. Magni sequi veniam, enim suscipit vitae libero eaque cupiditate ipsum nisi harum illum cum iure. Quod totam sequi rerum voluptatibus pariatur rem magnam natus tempora aliquid facere, cumque porro impedit repudiandae perspicatis, obcaecati eligendi magnam beatae, praesentium explicabo eaque quisquam odit deleniti pariatur, accusantium vitae nostrum vel esse? Voluptatem accusamus enim odit, tempora ipsam ipsa velit, explicabo excepturi vel placeat nisi provident reprehenderit exercitationem dolor voluptates cupiditate. Fugit eaque minus deserunt ab nulla, incidunt sint debitis labore magnam aspernatur mollitia suscipit blanditiis iusto corporis. Animi vel deserunt unde, velit iusto molestiae eligendi aliquid dolore, nemo dignissimos voluptatem voluptas laboriosam quo hic, architecto laudantium doloremque sunt asperiores suscipit. Exercitationem natus officiis at neque ipsum animi libero eaque laborum corrupti eveniet, itaque culpa sequi est tempora fugiat eum, assumenda animi itaque non sequi at magni nesciunt rerum esse minima, eius voluptate eos accusantium in delectus praesentium doloremque totam modi illo quis? Commodi officia repellat quis, nam eaque iste dolorum consectetur blanditiis dolore tempore, beatae maiores recusandae commodi aspernatur ipsa accusantium eius officia odio ut, ducimus eos molestiae a minus quod vitae quas? Enim ad laborum nemo mollitia corporis perferendis necessitatibus quam commodi est expedita, fugit enim ea consequuntur cum nobis incidunt ad doloremque odio omnis

quas.Dolorum totam quidem sit itaque cupiditate ducimus
dignissimos reiciendis laborum, adipisci nobis non voluptate
provident libero aut facilis ipsum quam sequi, aliquid